1 I

1.1 (1)

$$L = T - V = \frac{1}{2}mr^2\dot{\phi}^2.$$

1.2 (2)

Given the Lagrangian L, the conjugate momentum p_j corresponding to a generalized coordinate q_j is defined as

$$p_j = \frac{\partial L}{\partial \dot{q}_j}.$$

Thus,

$$p_\phi = \frac{\partial L}{\partial \dot{\phi}} = mr^2\dot{\phi}.$$

1.3 (3)

Using the Legendre transformation, the Hamiltonian is

$$\begin{aligned} H &= p_\phi\dot{\phi} - L \\ &= \frac{1}{2}mr^2\dot{\phi}^2 \\ &= \frac{p_\phi^2}{2mr^2}. \end{aligned}$$

Comparing this with the given form, we obtain

$$I = mr^2.$$

2 II

2.1 (4)

The time-independent Schrödinger equation (TISE) is

$$\hat{H}f(\phi) = Ef(\phi).$$

In the ϕ-representation, it becomes

$$-\frac{\hbar^2}{2I}\frac{d^2f(\phi)}{d\phi^2} = Ef(\phi).$$

The general solution is

$$f(\phi) = Ae^{ik\phi} + Be^{-ik\phi}, \qquad k = \frac{\sqrt{2IE}}{\hbar}.$$

Considering the periodic boundary condition

$$f(\phi + 2\pi) = f(\phi),$$

we obtain

$$e^{\pm i2\pi k} = 1,$$

which requires k to be an integer.

Therefore, the energy eigenvalues are

$$E_n = \frac{\hbar^2 n^2}{2I}, \qquad n = 0, 1, 2, \ldots$$

A normalized basis of energy eigenfunctions is

$$f_n(\phi) = \frac{1}{\sqrt{2\pi}} e^{in\phi}, \qquad n \in \mathbb{Z}.$$

For $n = 0$, the normalized eigenfunction is unique up to an arbitrary global phase. For $n \geq 1$, the energy level is two-fold degenerate, and any normalized linear combination of $e^{in\phi}$ and $e^{-in\phi}$ is also an energy eigenfunction.

3 III

3.1 (5)

Since

$$\vec{B} = \nabla \times \vec{A},$$

and both $\vec{A}$ and $\vec{B}$ have cylindrical symmetry, we can apply Stokes' theorem to a circular loop $\mathcal{C}_\rho$ of radius ρ, centered on the z-axis:

$$\oint_{\mathcal{C}_\rho} \vec{A} \cdot d\vec{l} = \int_{\mathcal{S}_\rho} \vec{B} \cdot d\vec{S}.$$

Since $\vec{A} = A_\phi(\rho)\vec{e}_\phi$, we have

$$2\pi\rho A_\phi(\rho) = \begin{cases} B\pi\rho^2, & \rho < \ell, \\ B\pi\ell^2, & \rho > \ell. \end{cases}$$

Thus,

$$A_\phi(\rho) = \begin{cases} \dfrac{B\rho}{2}, & \rho < \ell, \\ \dfrac{B\ell^2}{2\rho}, & \rho > \ell. \end{cases}$$

Since the particle moves on a circular orbit of radius $r > \ell$, we have

$$\vec{A}(r) = \frac{B\ell^2}{2r}\vec{e}_\phi.$$

Comparing this with the given form

$$\vec{A} = \frac{a}{r}\vec{e}_\phi,$$

we obtain

$$\boxed{a = \frac{B\ell^2}{2}}.$$

3.2 (6)

The Lagrangian in the presence of the magnetic field is

$$L^{(B)} = \frac{1}{2}mr^2\dot{\phi}^2 + q\vec{v}\cdot\vec{A}.$$

Since

$$\vec{v} = r\dot{\phi}\vec{e}_\phi, \qquad \vec{A} = \frac{B\ell^2}{2r}\vec{e}_\phi,$$

we obtain

$$L^{(B)} = \frac{1}{2}mr^2\dot{\phi}^2 + \frac{qB\ell^2}{2}\dot{\phi}.$$

The canonical momentum conjugate to ϕ is

$$p_\phi = \frac{\partial L^{(B)}}{\partial\dot{\phi}} = mr^2\dot{\phi} + \frac{qB\ell^2}{2}.$$

Using the Legendre transformation,

$$\begin{aligned} H^{(B)} &= p_\phi\dot{\phi} - L^{(B)} \\ &= \left(p_\phi - \frac{qB\ell^2}{2}\right)\dot{\phi} - \frac{1}{2}mr^2\dot{\phi}^2. \end{aligned}$$

From the definition of canonical momentum,

$$\dot{\phi} = \frac{1}{mr^2}\left(p_\phi - \frac{qB\ell^2}{2}\right).$$

Substituting this into the Hamiltonian, we obtain

$$\boxed{H^{(B)} = \frac{1}{2I}\left(p_\phi - \frac{qB\ell^2}{2}\right)^2}.$$

3.3 (7)

The TISE is

$$\hat{H}^{(B)}f(\phi) = E^{(B)}f(\phi).$$

Replacing the canonical momentum p_ϕ with $-i\hbar\partial/\partial\phi$, we obtain

$$\hat{H}^{(B)} = \frac{1}{2I}\left(-i\hbar\frac{\partial}{\partial\phi} - \frac{qB\ell^2}{2}\right)^2.$$

Substituting the eigenfunctions

$$f_n(\phi) = \frac{1}{\sqrt{2\pi}}e^{in\phi}$$

into the TISE, we obtain

$$\boxed{E_n^{(B)} = \frac{1}{2I}\left(n\hbar - \frac{qB\ell^2}{2}\right)^2}, \qquad n \in \mathbb{Z}.$$

3.4 (8)

Define

$$\eta = \frac{qB\ell^2}{2\hbar}.$$

The energy eigenvalues can then be written as

$$E_n^{(B)} = \frac{\hbar^2}{2I}(n-\eta)^2.$$

As B increases from zero, the ground state first becomes degenerate when the $n=0$ and $n=1$ energy levels become equal:

$$E_0^{(B)} = E_1^{(B)}.$$

Thus,

$$\eta^2 = (1-\eta)^2,$$

which gives

$$\eta = \frac{1}{2}.$$

Therefore,

$$\boxed{a = \frac{B\ell^2}{2} = \frac{\hbar}{2q}}.$$

4 IV

4.1 (9)

By definition,

$$\mathcal{P}f(\phi) = f(-\phi).$$

Applying the parity operator twice gives

$$\mathcal{P}^2 f(\phi) = f(\phi),$$

and hence

$$\mathcal{P}^2 = \mathbb{I}, \qquad \mathcal{P}^{-1} = \mathcal{P}.$$

Furthermore, using the inner product,

$$\begin{aligned}\langle \mathcal{P}f | \mathcal{P}g \rangle &= \int_{-\pi}^{\pi} f^*(-\phi) g(-\phi)\, d\phi \\ &= \int_{-\pi}^{\pi} f^*(\phi) g(\phi)\, d\phi \\ &= \langle f | g \rangle.\end{aligned}$$

Therefore, the parity operator is unitary:

$$\boxed{\mathcal{P}^\dagger = \mathcal{P}^{-1}}.$$

4.2 (10)

The Hamiltonian after introducing the electrostatic potential $V(\phi)$ is

$$\hat{H}^{(E)} = -\frac{\hbar^2}{2I}\frac{d^2}{d\phi^2} + V(\phi).$$

Applying the parity operator to $\hat{H}^{(E)} f(\phi)$, we obtain

$$\begin{aligned}\mathcal{P}\hat{H}^{(E)} f(\phi) &= (\hat{H}^{(E)} f)(-\phi) \\ &= -\frac{\hbar^2}{2I}\frac{d^2 f(-\phi)}{d\phi^2} + V(-\phi) f(-\phi).\end{aligned}$$

On the other hand,

$$\hat{H}^{(E)}\mathcal{P} f(\phi) = -\frac{\hbar^2}{2I}\frac{d^2 f(-\phi)}{d\phi^2} + V(\phi) f(-\phi).$$

Since $V(-\phi) = V(\phi)$, the two expressions are identical. Hence,

$$\boxed{[\mathcal{P}, \hat{H}^{(E)}] = 0}.$$

4.3 (11)

Before introducing $V(\phi)$, the first excited energy level is two-fold degenerate, with the normalized eigenfunctions

$$f_c(\phi) = \frac{1}{\sqrt{\pi}}\cos\phi, \qquad f_s(\phi) = \frac{1}{\sqrt{\pi}}\sin\phi.$$

Since $V(\phi)$ is even, the off-diagonal matrix element vanishes:

$$\langle f_c | V | f_s \rangle = \frac{1}{\pi}\int_{-\pi}^{\pi} V(\phi)\cos\phi\sin\phi\, d\phi = 0.$$

Therefore, the first-order energy corrections are

$$E_c^{(1)} = \frac{1}{\pi}\int_{-\pi}^{\pi} V(\phi)\cos^2\phi\, d\phi,$$

and

$$E_s^{(1)} = \frac{1}{\pi}\int_{-\pi}^{\pi} V(\phi)\sin^2\phi\, d\phi.$$

The energy difference between the two levels is

$$\begin{aligned}\Delta E &= \left| E_c^{(1)} - E_s^{(1)} \right| \\ &= \boxed{\frac{1}{\pi}\left|\int_{-\pi}^{\pi} V(\phi)\cos(2\phi)\, d\phi\right|}.\end{aligned}$$